

# Why the Icosahedron

Three, Five, and Non-Commutation

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# **Why the Icosahedron: Three, Five, and Non-Commutation**

### Abstract

FFGFT is a  $\mathbb{Z}_3$  theory from the ground up. Yet the lepton masses contain the golden ratio  $\phi$ , and  $\phi$  is the number of five. This document explains why the icosahedron is the bridge: its rotation group  $A_5$  is the smallest finite group in which a three and a five coexist without commuting. From this non-commutation arises a mixing of the  $\mathbb{Z}_3$  modes under the five-fold rotation, and the mixing has one rational component  $2/9$  and two  $\phi$ -bearing components. The rational one is the Koide phase (Doc. 367), the  $\phi$ -bearing ones are the skeleton of the mass ratios (Doc. 368). A single matrix element of this mixing carries the entire Koide formula. The document computes nothing new; it narrates what Docs. 285, 293, 367 and 368 mean together.

## Notation

### Epistemic markers:

|            |                                                           |
|------------|-----------------------------------------------------------|
| <b>[B]</b> | algebraically proved (check script in the cited document) |
| <b>[K]</b> | numerically confirmed                                     |
| <b>[S]</b> | structurally observed, derivation open                    |
| <b>[E]</b> | established external mathematics                          |

| Symbol              | Meaning                                                                              |
|---------------------|--------------------------------------------------------------------------------------|
| $\mathbb{Z}_3, C_3$ | cyclic group of order 3; $C_3$ : permutation $(x, y, z) \mapsto (y, z, x)$           |
| $v_0, v_1, v_2$     | Fourier modes, eigenvectors of $C_3$                                                 |
| $\omega$            | $e^{2\pi i/3}$                                                                       |
| $A_5$               | icosahedral group, 60 elements                                                       |
| $R_5$               | rotation by $72^\circ$ about the icosahedron vertex $(0, 1, \phi)$                   |
| $\phi$              | golden ratio $(1 + \sqrt{5})/2$                                                      |
| $p_0, p_1, p_2$     | redistribution weights of the electron mode under $R_5$ (Doc. 293)                   |
| $A$                 | $\langle v_0   R_5 v_e \rangle,  A  = \sqrt{2}/3$ (Doc. 367)                         |
| $D_4$               | lattice on $T^4$ ; $ \text{Aut}(D_4)  = 1152$ (Docs. 314, 364)                       |
| $\text{GF}(3^k)$    | finite field of characteristic 3; $\text{GF}(9) \ni \phi, \text{GF}(81) \ni \zeta_5$ |

## 1 The Observation

There are three charged leptons. Their masses stand in the ratio  $m_e : m_\mu : m_\tau \approx 1 : 206.8 : 3477.4$ , and since 1981 this ratio has been known to obey a strikingly simple formula, the Koide formula:

$$\sqrt{m_k} = \sqrt{M} \left( 1 + \sqrt{2} \cos \left( \theta + \frac{2\pi k}{3} \right) \right), \quad k = 0, 1, 2.$$

The scale  $M$  is a unit – it only says how large the masses are overall. The phase  $\theta$  is the actual information: it determines the ratios. Empirically  $\theta = 2/9$  to  $10^{-7}$  (Doc. 292). Why  $2/9$ ? Why not  $1/4$ , not  $0.3$ ? And why the coefficient  $\sqrt{2}$ ?

The answer leads through a geometric object that at first has nothing to do with leptons: the icosahedron. The rest of this document explains why.

## 2 Where FFGFT Comes From: The Three

FFGFT begins with a  $\mathbb{Z}_3$ . Three generations, triality, the circulant as mass operator. The mode space is  $\mathbb{C}^3$ ; its natural basis is the Fourier modes

$$v_k = \frac{1}{\sqrt{3}} (1, \omega^k, \omega^{2k}), \quad k = 0, 1, 2.$$

These are the eigenvectors of the cyclic permutation  $C_3$ . The mode  $v_0$  is the symmetric one;  $v_1$  and  $v_2$  carry the phases  $\omega$  and  $\omega^2$ . The electron sits in  $v_1$ , the muon in  $v_2$ , the tau in  $v_0$  – this is the assignment carried by the Koide index  $k$ .

What this structure alone does *not* deliver is a number. The  $\mathbb{Z}_3$  says there are three modes and how they transform under  $C_3$ . It says nothing about masses. The Koide formula needs two inputs – the coefficient  $\sqrt{2}$  and the phase  $\theta = 2/9$  – and neither is contained in the  $\mathbb{Z}_3$ .

### 3 Where $\phi$ Comes From: The Five

The golden ratio is the number of the pentagon:  $\cos 72^\circ = (\phi - 1)/2$ , and the diagonals of a regular pentagon stand to its sides in the ratio  $\phi$  [E]. There is no five in the  $\mathbb{Z}_3$  structure. If  $\phi$  appears in the lepton masses – Doc. 368 shows  $p_1/p_2 = \phi^8$  exactly [B] – a five must come from somewhere.

### 4 Why the Icosahedron: Where Three and Five Meet

The icosahedron has 20 triangular faces and 12 vertices; at each vertex five triangles meet. Its rotation group  $A_5$  has 60 elements: the identity, 15 rotations by  $180^\circ$  (edge midpoints), 20 rotations by  $120^\circ$  (face centres), 24 rotations by  $72^\circ$  or  $144^\circ$  (vertices) [E]. Three and five are both present.

That is not trivial. The cyclic group  $\mathbb{Z}_{15}$  also contains elements of order 3 and 5 – but there they commute. In  $A_5$  they *do not commute*: a rotation by  $120^\circ$  followed by one of  $72^\circ$  is not the same as the reverse order.  $A_5$  is the smallest group in which this is so.

Concretely: in standard position the vertices of the icosahedron are the cyclic permutations of  $(0, \pm 1, \pm \phi)$ . The 3-fold axes pass through face centres; one of them is  $(1, 1, 1)$ , and the rotation by  $120^\circ$  about this axis is *exactly* the permutation  $C_3$  of FFGFT. The  $\mathbb{Z}_3$  of FFGFT is, without reinterpretation, a subgroup of the icosahedral group (Doc. 285). The 5-fold axes pass through vertices; one vertex is  $(0, 1, \phi)$ , and the rotation  $R_5$  by  $72^\circ$  about this axis is the five-fold rotation used in Doc. 293.

### 5 Where the Icosahedron Lives: Unrolled, Not Compact

Up to here it sounds as if the icosahedron were simply *the* structure of FFGFT. It is not – and the corpus is very precise about this (Docs. 285, 314, 364). FFGFT has two pictures connected by a lossless coordinate transformation, and the icosahedron lives in only one of them.

**Compact: the torus  $T^4$  with the  $D_4$  lattice.** This is the original. The four-torus carries the  $D_4$  lattice, and its triality is the  $\mathbb{Z}_3$  of FFGFT – as a lattice operation, as a circulant on the mode orbits (Doc. 314). The winding numbers of the T0 ladder live here. The lattice knows the three. It knows *no* five:  $|\text{Aut}(D_4)| = 1152 = 2^7 \cdot 3^2$ , and  $5 \nmid 1152$ . No icosahedron acts on a  $D_4$  lattice (Doc. 364, incompatibility theorem) [B].

**Unrolled:  $\mathbb{R}^3 \times S^1$ .** Observed space arises when one circle of the torus becomes time through the measurement process:  $T^4 \rightarrow \mathbb{R}^3 \times S^1$  (Doc. 285). On *this*  $\mathbb{R}^3$ ,  $A_5$  acts as rotation group. The  $C_3 \subset A_5$  is the same three as the triality in the compact picture – the rotation by  $120^\circ$  about  $(1, 1, 1)$ . But the five exists only here. The icosahedron is the image of the torus geometry in unrolled space, not the torus itself.

**How the five nevertheless enters the field.** Doc. 364 resolves the apparent tension: what matters is *as what* a symmetry enters – as an automorphism of lattice vectors or as a phase. The  $72^\circ$  rotation has characteristic polynomial  $(x - 1)(x^2 - (\phi - 1)x + 1)$ ; its trace is  $\phi$ . Modulo 3,  $\phi$  lies in  $\text{GF}(9)$  and is primitive there ( $\phi^4 = -1$ , order 8). The roots of the quadratic factor lie not in  $\text{GF}(9)$  but in  $\text{GF}(81)$ , and have order 5 there – because  $5 \mid 80 = |\text{GF}(81)^*|$  [B]. In the lattice, the  $72^\circ$  rotation is a vector automorphism, and  $5 \nmid 1152$ . In the field, the same rotation is only

its eigenvalue  $\zeta_5$ , and  $5 \mid 80$ . Geometric incompatibility becomes arithmetic compatibility as soon as one passes from vectors to phases. Doc. 364 verbatim: this is the mechanism behind Doc. 293 –  $\theta = 2/9$  arises from  $R_5$  as a phase, not as a lattice operation.

**Why this is forced.** In characteristic 3,  $x^3 - 1 = (x - 1)^3$  **[E]**: in no  $\text{GF}(3^k)$  is there a primitive cube root of unity. The three *cannot* be a phase and must be lattice. The five *cannot* be lattice ( $5 \nmid 1152$ ) and must be phase. The prime 3 distributes the two symmetries onto two complementary channels – not a choice, but a constraint. The extension  $\text{GF}(9) \rightarrow \text{GF}(81)$  that carries  $\zeta_5$  and  $\phi$  is precisely the doubling  $6 = 3 \oplus 3'$  of Doc. 285.

**What this means for the narrative.** When non-commutation of three and five is spoken of below, this is meant: the three as lattice operation in the compact picture, the five as phase in the unrolled one. They do not meet in geometry – there they are incompatible – but in the field  $\text{GF}(81)$ . The icosahedron is the visible form of this meeting.

## 6 What Non-Commutation Produces

Because  $R_5$  does not commute with  $C_3$ ,  $R_5$  is *not diagonal* in the mode basis  $v_0, v_1, v_2$ . Applying  $R_5$  to the electron mode  $v_1$ , the result does not land back in  $v_1$  but is distributed over all three modes. The moduli squared of the projections are the redistribution weights (Doc. 293) **[B]**:

$$p_0 = |\langle v_0 | R_5 | v_1 \rangle|^2 = \frac{2}{9}, \quad p_1 = |\langle v_1 | R_5 | v_1 \rangle|^2 = \frac{2 + 3\phi}{9}, \quad p_2 = |\langle v_2 | R_5 | v_1 \rangle|^2 = \frac{5 - 3\phi}{9}.$$

One sees at once:  $p_0$  is rational,  $p_1$  and  $p_2$  contain  $\phi$ . That is structure, not accident. The projection onto the symmetric mode  $v_0$  is blind to the five-fold structure – it only counts how much of the  $\mathbb{Z}_3$  mixing arrives, and that is  $2/9 = 2/3^2$ . The projections onto  $v_1$  and  $v_2$  see the five and therefore carry  $\phi$ .

It becomes even clearer at the symmetric mode itself. What  $R_5$  makes of  $v_0$ :

$$\left( |\langle v_0 | R_5 | v_0 \rangle|^2, |\langle v_0 | R_5 | v_1 \rangle|^2, |\langle v_0 | R_5 | v_2 \rangle|^2 \right) = \left( \frac{5}{9}, \frac{2}{9}, \frac{2}{9} \right) \quad \mathbf{[K]}$$

The complete matrix of moduli squared **[K]**:

| $ \langle v_j   R_5   v_k \rangle ^2$ | $k = 0$ | $k = 1$         | $k = 2$         |
|---------------------------------------|---------|-----------------|-----------------|
| $j = 0$                               | 5/9     | 2/9             | 2/9             |
| $j = 1$                               | 2/9     | $(2 + 3\phi)/9$ | $(5 - 3\phi)/9$ |
| $j = 2$                               | 2/9     | $(5 - 3\phi)/9$ | $(2 + 3\phi)/9$ |

Two things are immediately visible. Wherever  $v_0$  is involved, the entry is  $2/9$  or  $5/9$  – rational, without  $\phi$ . And  $\phi$  appears only between  $v_1$  and  $v_2$ , the two modes that see the five. Every row and every column sums to 1.

The amplitudes of the first row are  $(\sqrt{5}, \sqrt{2}, \sqrt{2})/3$ . The five-fold rotation leaves  $\sqrt{5}/3$  of the symmetric mode in itself and passes  $\sqrt{2}/3$  to each of the two non-trivial modes. And  $5 + 2 + 2 = 9 = 3^2$ . This is the one row in which the five of the icosahedron and the three of FFGFT stand side by side. The trace of  $R_5$  is  $1 + 2 \cos 72^\circ = \phi$  **[E]** – the five is in the operator, the three in the basis, and the product of both is the row  $(5, 2, 2)/9$ .

## 7 Doc. 367: Why $\theta = p_0$ , and How the Whole Koide Formula Follows

Doc. 293 had noted that  $p_0 = 2/9$  coincides with the Koide angle  $\theta = 2/9$ , and left the question of the reason open. Doc. 367 gives the answer: both are the same quotient  $2/3^2$ , read in different roles. Numerator: the non-trivial  $\mathbb{Z}_3$  modes. Denominator: the order squared, the dimension of the operator space  $\mathbb{C}^3 \otimes \mathbb{C}^3$ .

The decisive step comes next. The single matrix element  $A = \langle v_0 | R_5 v_e \rangle$  has modulus  $\sqrt{2}/3$  and modulus squared  $2/9$ . The Koide formula

$$a_k = 1 + \sqrt{2} \cos\left(\theta + \frac{2\pi k}{3}\right)$$

needs exactly these two numbers:  $\sqrt{2}$  as coefficient,  $2/9$  as phase. With  $3|A| = \sqrt{2}$  and  $|A|^2 = 2/9$  this becomes **[B]**:

$$a_k = 1 + 3|A| \cos\left(|A|^2 + \frac{2\pi k}{3}\right).$$

No free parameter. The modulus gives the amplitude, the modulus squared the phase – Born-rule structure, but here not as interpretation, rather as algebra: the same complex number, once as  $|A|$ , once as  $|A|^2$ . The Koide formula is thereby fully derived from the question: how much of the five-fold rotation lands in the symmetric  $\mathbb{Z}_3$  mode?

## 8 Doc. 368: The $\phi$ -Skeleton

The two other weights carry the five. Their ratios are pure  $\phi$ -powers **[B]**:

$$\frac{p_1}{p_2} = \phi^8, \quad \frac{p_0}{p_2} = 2\phi^4, \quad 3\sqrt{p_j} \in \{\sqrt{2}, \phi^2, \phi^{-2}\}.$$

And  $\phi^8 \approx 46.98$  is, up to a Galois factor, the ratio  $m_\tau/m_e$ :  $74\phi^8$  hits to 0.03 %, with  $\xi$  correction to 0.003 % **[K]**. That is the second role of the icosahedron:  $p_0$  supplies the phase,  $p_1$  and  $p_2$  supply the skeleton of the mass ratios. Both from the same five-fold rotation, both without  $\phi$  having been inserted anywhere.

## 9 Doc. 293: Why It Is the Icosahedron and Nothing Else

Doc. 293 tested whether  $2/9$  depends on the axis. Two hundred random axes with angle  $72^\circ$ : none yields  $2/9$ . Angle scan at fixed icosahedral axis: only  $72^\circ$  yields  $2/9$ . Order series – rotations of order 2, 3, 4, 5, 6 about icosahedral axes: only order 5 yields  $2/9$  **[K]**. The number is icosahedron-specific. It arises because the axis  $(0, 1, \phi)$  and the angle  $72^\circ$  together stand to the  $(1, 1, 1)$  axis of the  $\mathbb{Z}_3$  exactly as the icosahedral geometry prescribes.

## 10 What the Icosahedral $\mathbb{Z}_3$ Is – and What It Is Not

A comparison sharpens the concept. In string theory there are orbifolds  $T^6/\mathbb{Z}_3$  whose  $\mathbb{Z}_3$  acts as  $\omega \cdot \mathbb{1}$  on  $\mathbb{C}^3$  – all three coordinates receive the same phase. This  $\mathbb{Z}_3$  is *central*: it commutes with every matrix. The icosahedral  $\mathbb{Z}_3$  is  $C_3$  with eigenvalues  $(1, \omega, \omega^2)$ , trace zero, *not central*.

Both structures count  $2/3^2$  – the orbifold CFT via the sum of twist-field weights  $1/9 + 1/9$  **[E]** (Doc. 367, external parallel). But only the icosahedral  $\mathbb{Z}_3$  can be *mixed* by  $R_5$ . For the central one,  $\langle v_0 | R_5 | v_1 \rangle$  would vanish, because  $R_5$  would then commute with  $\mathbb{Z}_3$  and

preserve sectors. The non-commutation that produced the number above is exactly what distinguishes the icosahedral structure from an orbifold.  $A_5$  has trivial centre; no element of  $A_5$  can become  $\omega \cdot 1$  in a faithful representation. The two  $\mathbb{Z}_3$ 's are different objects that happen – or do not happen – to share the same counting quantity.

## 11 What Remains Open

Two questions raised by this document have since been answered – one negatively, one by reversal. What remains after that is a single one.

**The Galois factors: the icosahedron does not explain them.**  $\phi^8$  is exact, but  $m_\tau/m_e$  is  $74\phi^8$  with  $74 = 2 \cdot 37$ , and 37 is the number that appears in  $K_{\text{frak}}$  and in  $1/\alpha = 3700/27$  (Doc. 338). In the phase channel (Sect. 5), 37 is the root of unity  $\zeta_{37}$ , and it lives in  $\text{GF}(3^{18})$ : the order of 3 modulo 37 is 18. The icosahedral phase  $\zeta_5$  lives in  $\text{GF}(81) = \text{GF}(3^4)$ . And  $4 \nmid 18$ :  $\text{GF}(81)$  is *not* a subfield of  $\text{GF}(3^{18})$  **[B]**. The two phases sit in incompatible extensions. That is a negative result: 37 does not come from the icosahedron, neither geometrically nor via the phase channel. What  $\text{GF}(3^{18})$  does contain is  $\phi$  (from  $\text{GF}(9)$ ) and the lepton order 26 (from  $\text{GF}(729)$ ); its “new” primes, absent from every subfield, are 19 and 37. The degree  $18 = 2 \cdot 3^2$  is the degree of  $\phi$  times  $|\mathbb{Z}_3|^2$  – the compositum of the  $\phi$ -extension with the  $\mathbb{Z}_3^2$ -extension  $\text{GF}(3^9)$ . Why this compositum, and why 37 rather than 19, is open **[S]**.

**The fractal correction: derived in Reading B.** Doc. 352 takes the T0 ladder as fundamental; then the step from percent to  $10^{-3}$  level is a measured correction, and its origin is attributed to QED (Reading A). This document sets Koide as fundamental, because it follows from the icosahedron. In this Reading B, the T0 ladder  $r_k \xi^{p_k}$  is a rational approximation to the transcendental amplitudes  $a_k^2$ , and the correction is the distance between cosine and power law – not measured but computed **[K]**. It differs for the three leptons because the three cosine values differ. Both readings give the same numbers; Doc. 369 § 4 places them side by side, Register R121 records the change of status. What remains open in Reading B is not the correction but: why are the approximation parameters exactly the  $\text{GF}(3^n)$  winding numbers, and why is the approximation so good?

**The one question that remains.** Both points converge on the same thing. The icosahedron supplies phases in  $\text{GF}(81)$ :  $\zeta_5$ ,  $\phi$ , the Koide angle. Number theory supplies orders in  $\text{GF}(3^n)$ : winding numbers, 37,  $K_{\text{frak}}$ . Both sides hit the lepton masses – the one parameter-free to  $10^{-3}\%$  as ratios, the other with an anchor at percent level as absolute numbers. But they do not meet:  $\text{GF}(81)$  and  $\text{GF}(3^{18})$  share only  $\text{GF}(9)$ , hence  $\phi$  and nothing else. Whether there is a structure in which the icosahedral phase and the Galois order have the same root – or whether they are two independent approaches that touch in  $\phi$  and otherwise remain separate – is the open edge of this document **[S]**.

## 12 The Short Version

The icosahedron is the place where three and five do not get along. FFGFT brings the three; the icosahedron is the smallest structure that adds a five without the five commuting with the three. From this non-commutation arises a mixing, and the mixing has one rational component –  $2/9$  – and two  $\phi$ -components. The rational one is the Koide phase. The  $\phi$ -components are the skeleton of the masses. A single matrix element of this mixing carries the entire Koide formula, parameter-free.